

MCMURDO STATION, Antarctica

WMO: 896640

Lat: 77.850S

Lon: 166.667E

Elev: 79

StdP: 14.65

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-33.6	-30.5	-44.3	0.4	-31.7	-41.1	0.5	-28.5	47.7	0.8	39.4	-5.8	8.3	80	1.231

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	37.4	30.6	35.3	29.7	33.5	28.7	32.0	36.0	30.5	34.2	29.2	32.9	6.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.8	21.9	32.5	26.2	20.2	31.8	24.5	18.6	30.8	11.8	36.3	11.2	34.3	10.7	33.2	39.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.2	29.3	25.0	-38.5	41.2	5.8	3.3	-42.7	43.6	-46.1	45.5	-49.4	47.3	-53.6	49.6		
(5)			-38.2	34.4	4.3	2.9	-41.3	36.4	-43.8	38.1	-46.2	39.8	-49.3	41.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	27.2	16.8	2.2	-5.0	-6.1	-9.7	-12.4	-13.0	-10.0	1.2	16.5	27.2
	DBStd	16.98	4.82	7.05	8.29	8.79	10.61	10.34	10.65	11.47	10.07	7.86	7.54	4.79
	HDD50	17206	706	929	1481	1651	1738	1790	1933	1954	1800	1513	1004	707
	HDD65	22681	1171	1349	1946	2101	2203	2240	2398	2419	2250	1978	1454	1172
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.8	9.0	11.4	12.1	10.7	11.0	11.7	10.8	10.6	11.0	11.3	10.6	9.3
Precipitation	PrecAvg	6.9	0.6	0.9	0.6	0.6	0.9	1.0	0.6	0.5	0.4	0.6	0.5	0.4
	PrecMax	13.5	2.8	2.0	2.4	1.6	3.1	5.3	2.7	1.9	1.4	2.2	1.7	0.9
	PrecMin	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.8	0.7	0.6	0.5	0.6	0.8	1.2	0.6	0.5	0.3	0.5	0.5	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	33.9	22.7	17.7	20.5	18.9	17.7	16.2	18.5	22.4	34.0	40.4
		MCWB	33.9	27.6	19.4	14.9	18.1	16.1	16.4	13.6	15.7	19.2	27.7	33.1
	2%	DB	37.7	30.1	19.4	14.6	15.4	13.0	12.5	11.9	13.3	17.8	31.0	37.6
		MCWB	31.3	25.7	16.8	12.7	13.3	11.2	10.4	9.9	11.0	14.4	25.7	30.3
	5%	DB	35.5	28.0	16.6	11.1	12.3	9.9	8.3	8.2	9.5	14.6	28.7	35.7
		MCWB	30.1	24.0	14.2	9.1	10.2	8.0	6.6	6.4	7.5	12.0	24.3	29.6
	10%	DB	33.8	26.0	14.0	7.5	9.0	6.2	3.6	5.0	5.3	12.0	26.5	34.0
		MCWB	29.2	22.2	11.8	5.9	7.2	4.5	2.1	3.8	3.5	9.9	22.5	28.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	28.9	19.9	15.1	18.5	15.6	16.5	14.1	14.4	18.7	28.1	34.0
		MCDB	40.1	31.8	22.6	16.7	20.6	17.9	18.2	15.9	16.3	21.7	31.4	39.6
	2%	WB	32.6	26.0	16.8	12.9	13.8	11.3	9.6	10.3	10.1	14.0	26.3	31.5
		MCDB	37.0	29.4	19.2	14.8	15.5	12.6	11.2	11.8	12.0	16.6	29.8	35.8
	5%	WB	31.1	24.0	14.2	9.2	10.3	8.0	5.6	6.7	7.0	11.7	24.2	30.3
		MCDB	34.8	27.5	16.3	11.1	12.1	9.7	7.4	8.1	8.8	13.8	27.7	34.4
	10%	WB	29.8	22.2	11.9	5.9	7.3	4.3	1.5	3.8	3.3	9.4	22.2	29.1
		MCDB	33.2	25.4	14.0	7.5	9.2	5.9	2.9	5.2	4.9	11.5	26.0	33.3
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.3	8.7	9.0	9.5	9.7	10.0	10.4	11.2	10.3	9.2	7.4
		MCDBR	9.6	10.7	9.6	12.8	11.6	12.7	12.9	12.0	14.6	11.8	9.4	9.6
		MCWBR	6.8	8.7	8.5	12.0	10.5	11.8	12.5	11.1	13.4	10.5	7.7	6.5
	5% WB	MCDBR	8.8	10.3	9.4	12.9	10.8	12.7	13.8	11.6	13.9	11.9	9.5	8.7
		MCWBR	6.9	8.6	8.5	12.1	10.1	11.9	12.9	11.1	13.0	10.7	7.9	6.4
Clear-Sky Solar Irradiance	taub	0.236	0.218	0.174	0.113	N/A	N/A	N/A	0.111	0.165	0.196	0.217	0.234	
	taud	2.248	2.262	2.177	1.896	N/A	N/A	N/A	1.877	2.126	2.235	2.256	2.246	
	Ebn at Noon	306	285	243	36	0	0	0	30	242	294	313	314	
	Edh at Noon	31	24	15	1	0	0	0	1	15	24	30	33	
All-Sky Solar Radiation	RadAvg	2536	1464	484	42	0	0	0	6	272	1128	2333	2952	
	RadStd	69	59	21	2	0	0	0	1	16	42	70	87	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page